

Coffee Machine Hardware Modification Guide

How to replicate the physical modifications of this build

1. Before Starting

Use this guide only with the same coffee-machine model and the same hardware revision used for the original build. Before soldering, compare the board with the reference photographs.

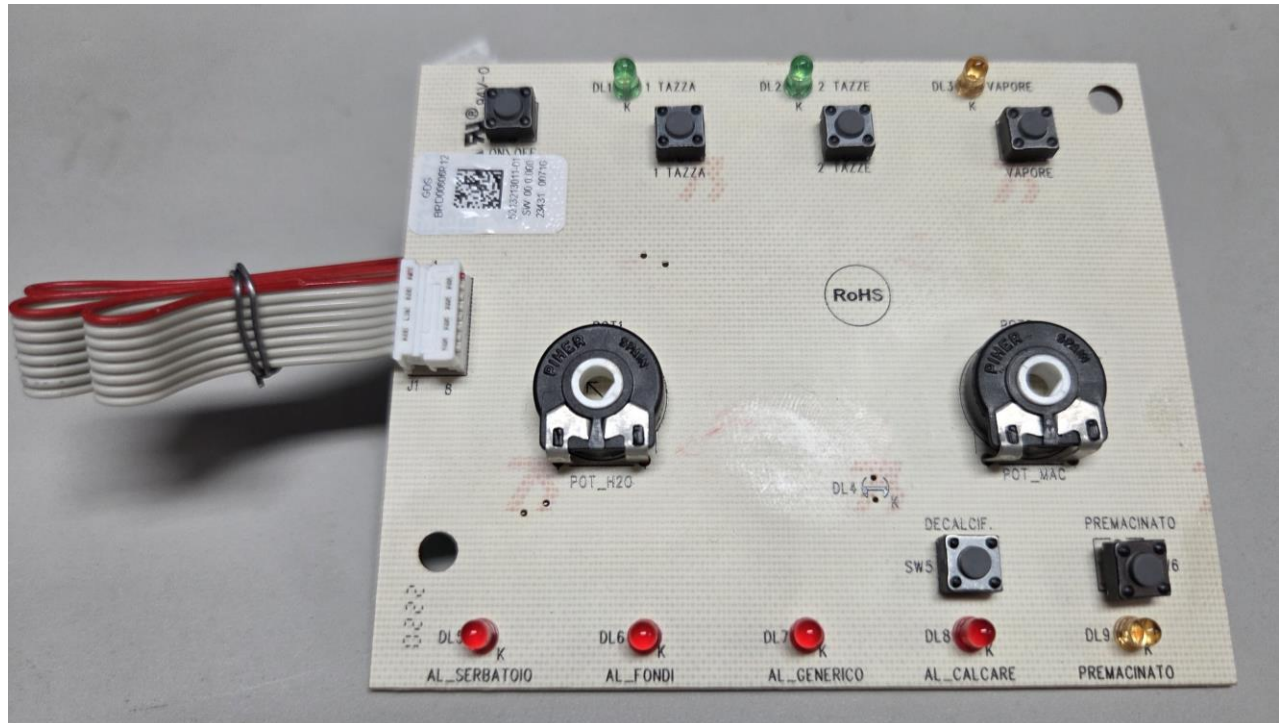
- Disconnect the machine from mains power before opening it.
- Allow heaters and water/steam components to cool.
- Do not connect controller GPIO directly to unknown coffee-machine signals. Follow the proven optocoupler/interface schematic.

2. Parts Required

Part	Type / Value	Quantity	Notes
De'Longhi Magnifica ESAM3200	coffee machine	1	The control board requires modification before installing the system.
PC817 8 Kanal	Isolationsmodul	2	Both modules require modification before installation.
PC817 4 Kanal	Isolationsmodul	1	
ISO1540	I2C Isolator	1	
LOLIN32 Lite	Microcontroller	1	
PCF8563	Real-Time Clock	1	
GL-A22	Ultrasonic Distance Sensor	1	
PCA9546 (4-Channel)	I2C Multiplexer	1	
MCP23017 (16-Bit)	I2C I/O Expander	1	
E18-D80NK	Infrared Object Detection Sensor	1	
MCP4018	Digital Potentiometer	2	
ADS1115	Analog-to-Digital Converter	1	
AC-DC Power Supply Module	5V	1	Connects to the BAT connector on the LOLIN32 Lite.

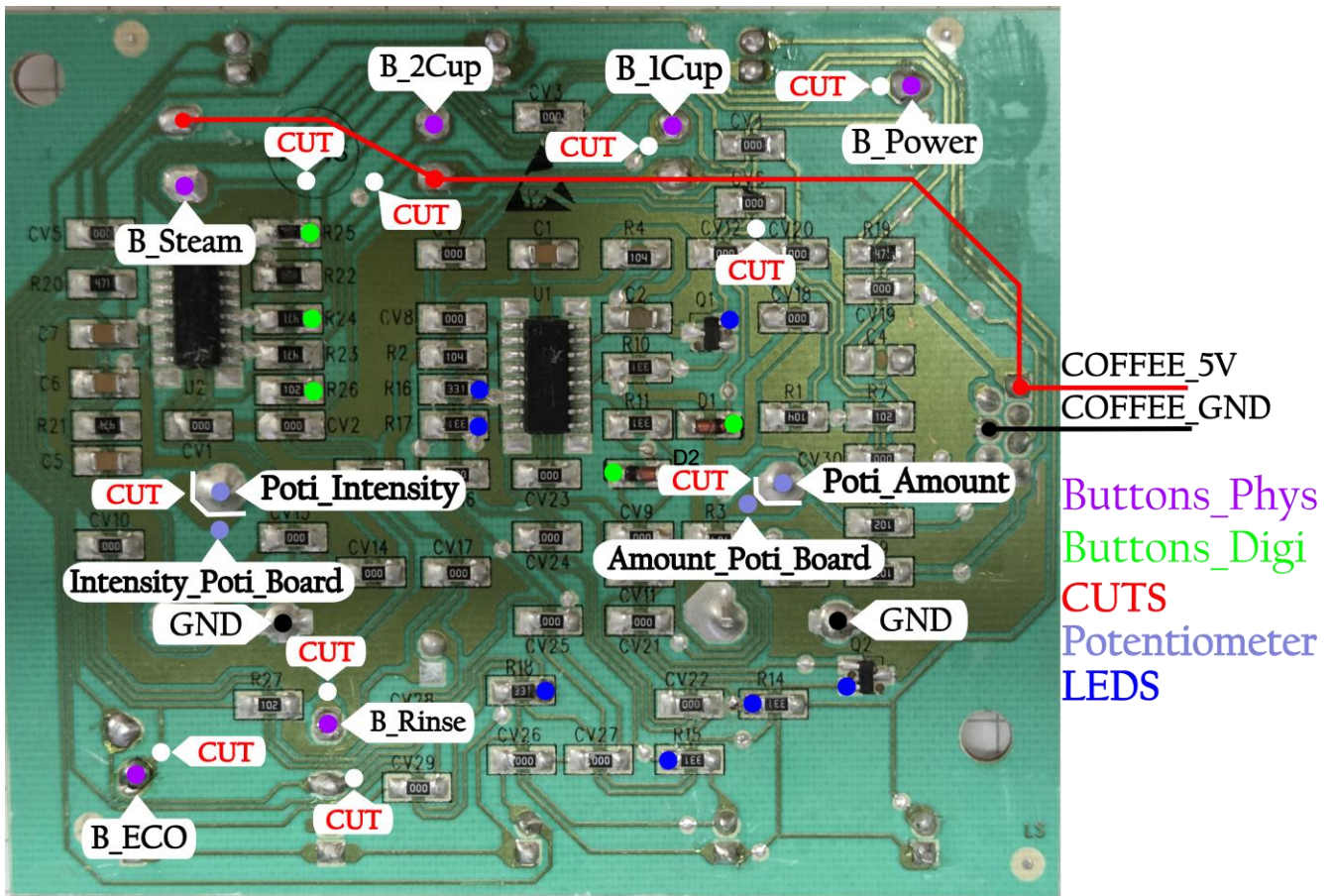
3. Access the Control Board

1. Disconnect the coffee machine from mains power.
2. Remove the housing panels required to access the control board.
3. Locate the control board shown in the reference photo below.
4. Remove or position the board so the required solder points are accessible.
5. The control board is covered with a protective layer of hot glue. **Carefully remove the hot glue completely**, ensuring that no PCB traces or components are damaged in the process.



4. Modify the Coffee-Machine Control Board

ID	Machine Function	Notes	Connect to
B_1Cup	Dispense 1 Cup of coffee		4 Channel PC817 module
B_2Cup	Dispense 2 Cups of coffee		4 Channel PC817 module
B_Power	Machine ON/OFF		4 Channel PC817 module
B_Steam	Steam mode ON/OFF		4 Channel PC817 module
B_Rinse	Rinse		G3VM-61A1 PhotoMOS 2 channel side of module
B_ECO	ECO ON/OFF		G3VM-61A1 PhotoMOS 2 channel side of module
Poti_Intensity	Potentiometer Wiper for Intensity		Connect to ADS1115 input
Intensity_Poti_Board	PCB trace for Intensity		Connect to digital potentiometer for Intensity.
Poti_Amount	Potentiometer Wiper for Amount		Connect to ADS1115 input
Amount_Poti_Board	PCB trace for Amount		Connect to digital potentiometer for Amount.
Q1	LED Ground Q1	3 Channel side	3/5 PC817 module
Q2	LED Ground Q2	5 Channel side	3/5 PC817 module
R14	1 Cup LED		3/5 PC817 module
R15	2 Cup LED		3/5 PC817 module
R16	Steam LED		3/5 PC817 module
R17	Rinse LED		3/5 PC817 module
R18	ECO LED		3/5 PC817 module
CUT	PCB Trace Cut Points	Carefully cut PCB traces marked in the picture below. Make sure each trace is completely interrupted without damaging other traces or components.	
COFFEE_5V		After cutting PCB traces, connect the 5 V supply from the control board to the buttons , as shown in the picture below.	
D1	Button Press Simulation	Buttons are multiplexed. Connect according to the schematic.	G3VM-61A1 PhotoMOS 6 channel module side
D2	Button Press Simulation	Buttons are multiplexed. Connect according to the schematic.	G3VM-61A1 PhotoMOS 6 channel module side
R24	Button Press Simulation	Buttons are multiplexed. Connect according to the schematic.	G3VM-61A1 PhotoMOS 6 channel module side
R25	Button Press Simulation	Buttons are multiplexed. Connect according to the schematic.	G3VM-61A1 PhotoMOS 6 channel module side
R26	Button Press Simulation	Buttons are multiplexed. Connect according to the schematic.	G3VM-61A1 PhotoMOS 6 channel module



4.1 Soldering Procedure

1. Cut the PCB traces at the locations indicated in the annotated picture.
2. Solder the **COFFEE_5V** and **COFFEE_GND** wires to the specified points on the PCB.
3. Solder the wires for all remaining connections according to the annotated picture.
4. Keep stripped wire ends as short as possible and ensure that no solder bridges are created between adjacent pads or traces.
5. Provide sufficient strain relief so that movement of the wires does not put mechanical stress on the PCB pads.
6. Secure the soldered wires to the PCB using hot glue. This provides additional strain relief and helps prevent accidental pulling on the wires from damaging the PCB or solder pads.
7. Carefully inspect every solder joint, wire connection, and modified trace before proceeding to the next step.

5. Prepare the Optocoupler Modules

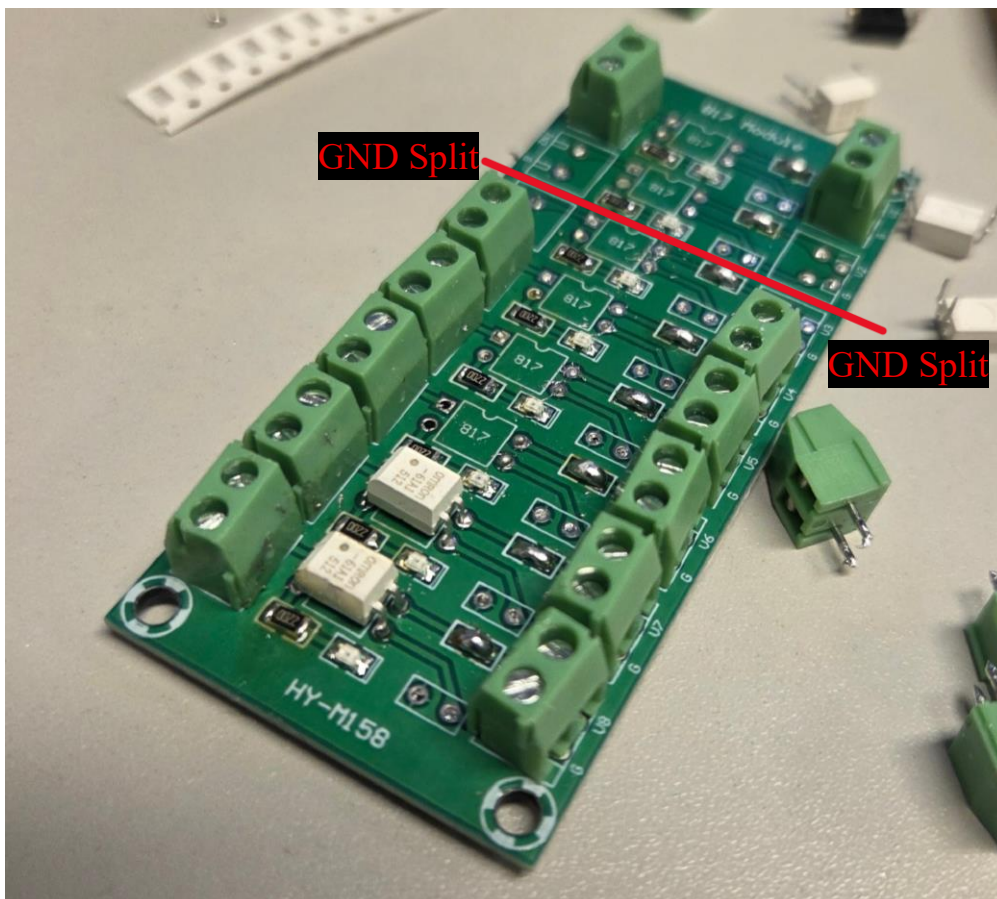
The modules used for **button simulation**, **physical button inputs**, and **LED detection** require **different modifications** before they can be installed.

5.1 Button Control – PhotoMOS Modification

For controlling the coffee-machine buttons, **G3VM-61A1 PhotoMOS relays** are used instead of the original **PC817 optocouplers** on the 8-channel module. The module must be modified by cutting the common GND traces on both sides of the board.

The required channels on the 8-channel PC817 module must be modified as follows:

1. Split the common GND of the module into two electrically isolated groups
 - 2 channels for use as inputs for the physical button switches, working together with the 4-channel PC817 module.
 - 6 channels for simulating button presses using the PhotoMOS relays.
2. Remove the **PC817** optocouplers from the required channels.
3. Replace them with **G3VM-61A1 PhotoMOS relays**.
4. Replace the original input resistors on these channels with **200 Ω resistors**.
5. Verify the orientation and connections of the G3VM-61A1 PhotoMOS relays before powering the board.



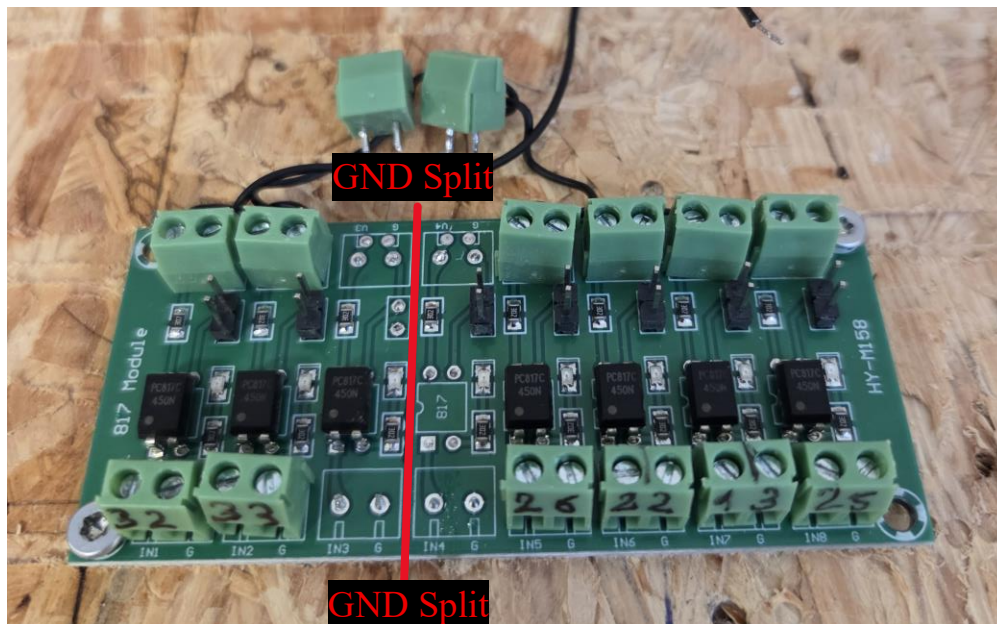
5.2 LED Module

The 8-channel module used for LED detection must be modified by **cutting the common GND traces on both sides of the board**.

The common GND must be split into two electrically isolated groups:

- **3 channels**
- **5 channels**

Cut the GND traces at the location shown in the picture below on **both** sides of the Board.



5.4 Physical Button Input Module

The 4-channel PC817 module is used together with the two input channels of the modified 8-channel module to detect the **physical button switches**. The **4-channel module does not require any modification** and can be used as supplied.

5.5 Check the Modifications

Before connecting the modules to the coffee-machine control board or controller:

1. Use a multimeter in continuity mode to verify that the separated GND sections are **no longer electrically connected**.
2. Check that no unintended PCB traces were damaged while cutting the GND traces.
3. Verify that all **G3VM-61A1 PhotoMOS relays** are installed correctly.
4. Verify that the required input resistors have been replaced with **200 Ω resistors**.
5. Compare all modifications with the reference pictures and final schematic before applying power.

8. Final Schematic

Finally, connect all components according to the **final schematic**.

To simplify the wiring, two custom PCBs were designed:

Custom PCB 1 – Controller Board

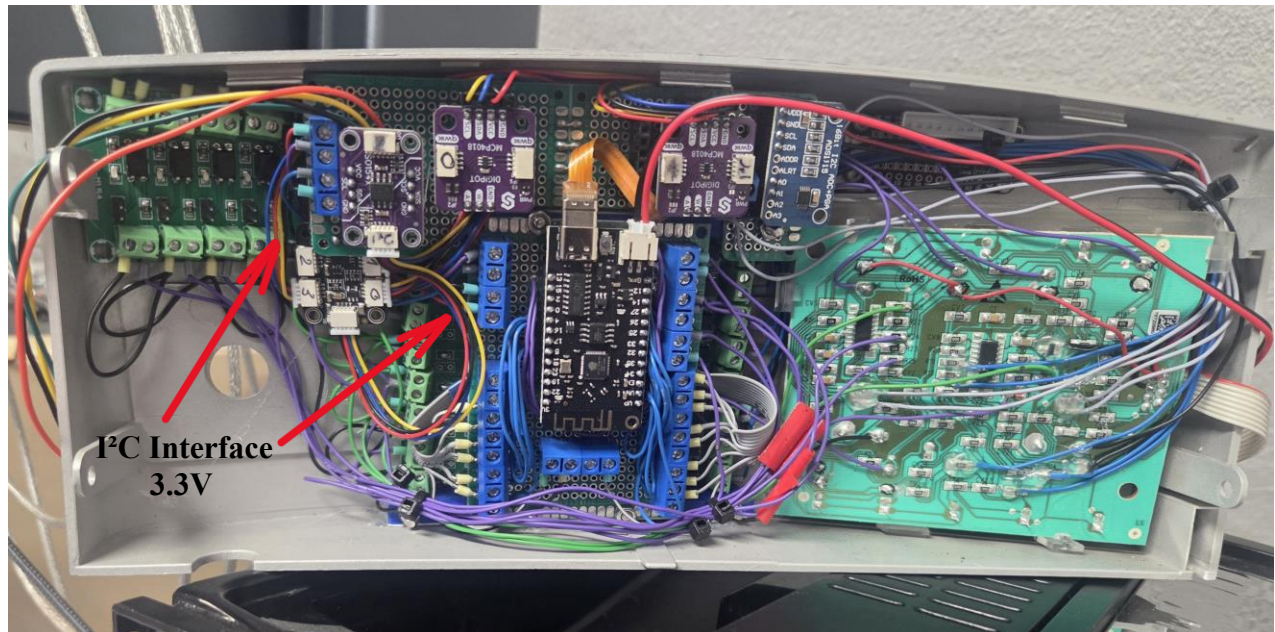
This PCB is designed for the **LOLIN32 Lite controller board** and expands the required pins to terminal blocks for easier wiring. The **PCF module** is also soldered directly onto this PCB and connected to the controller via **I²C**. Both the custom PCB and the PCF module are powered by the **3.3V supply from the LOLIN32 Lite**.

Custom PCB 2 – ADC and Digital Potentiometers

The second custom PCB is used for the **digital potentiometers, ADC, and isolated I²C interface**.

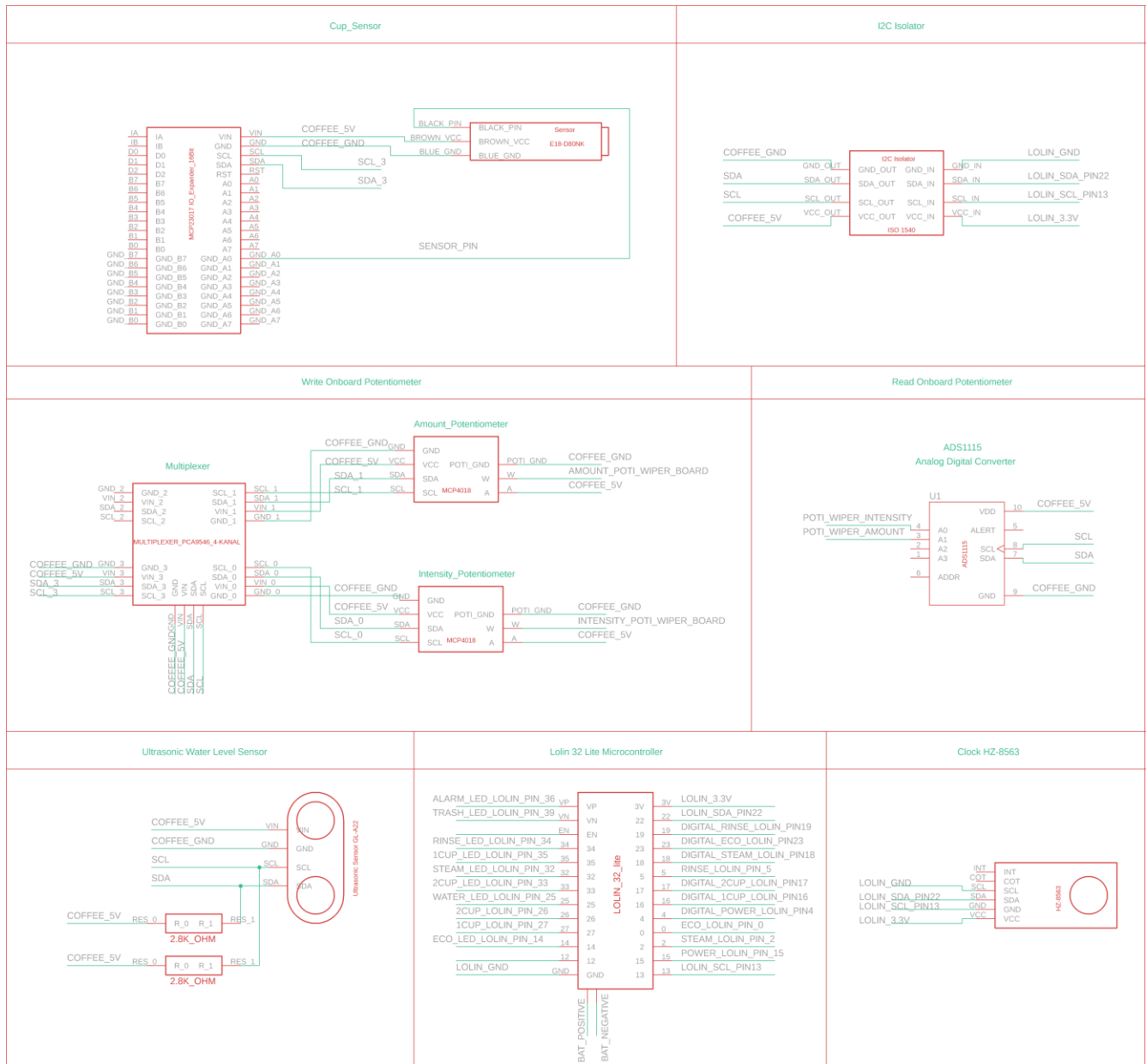
The board mainly provides easier connections by routing the required input and output signals to terminal blocks. The **I²C, VCC, and GND connections** of the individual modules are connected together as shown in the schematic. The board and its connected modules are powered by the **5 V supply from the coffee-machine control board**.

Connect all terminal blocks and remaining components according to the final schematic. Where possible, Qwiic connectors were used to simplify the I²C connections and reduce additional wiring.

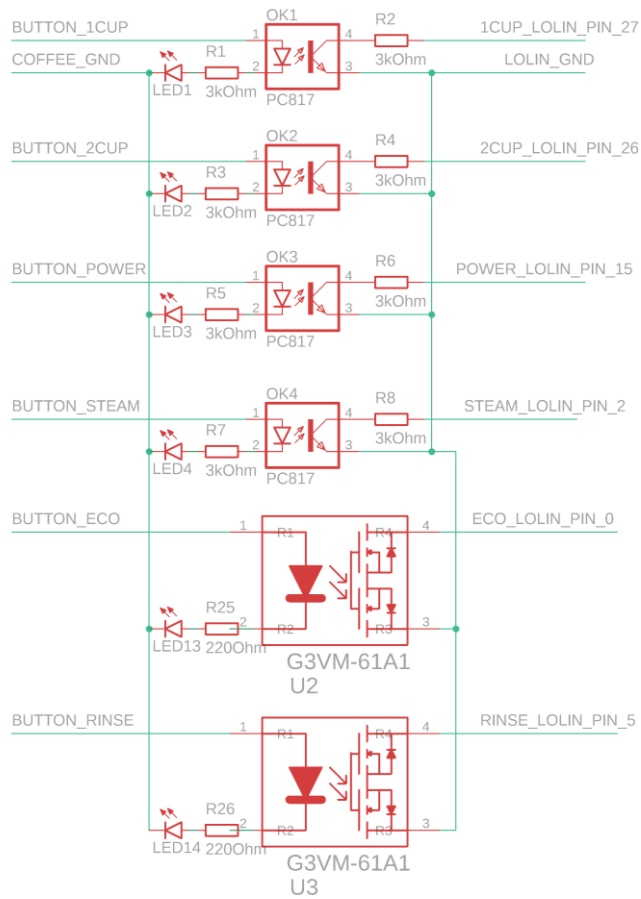


Important: Wire colors in the final implementation do not always match the schematic or documentation. The system was developed and modified throughout the build process, and replacing the existing wiring afterward would have introduced an unnecessary risk of damaging the completed system.

Always follow the **connection labels and final schematic** rather than the wire colors in the picture above.



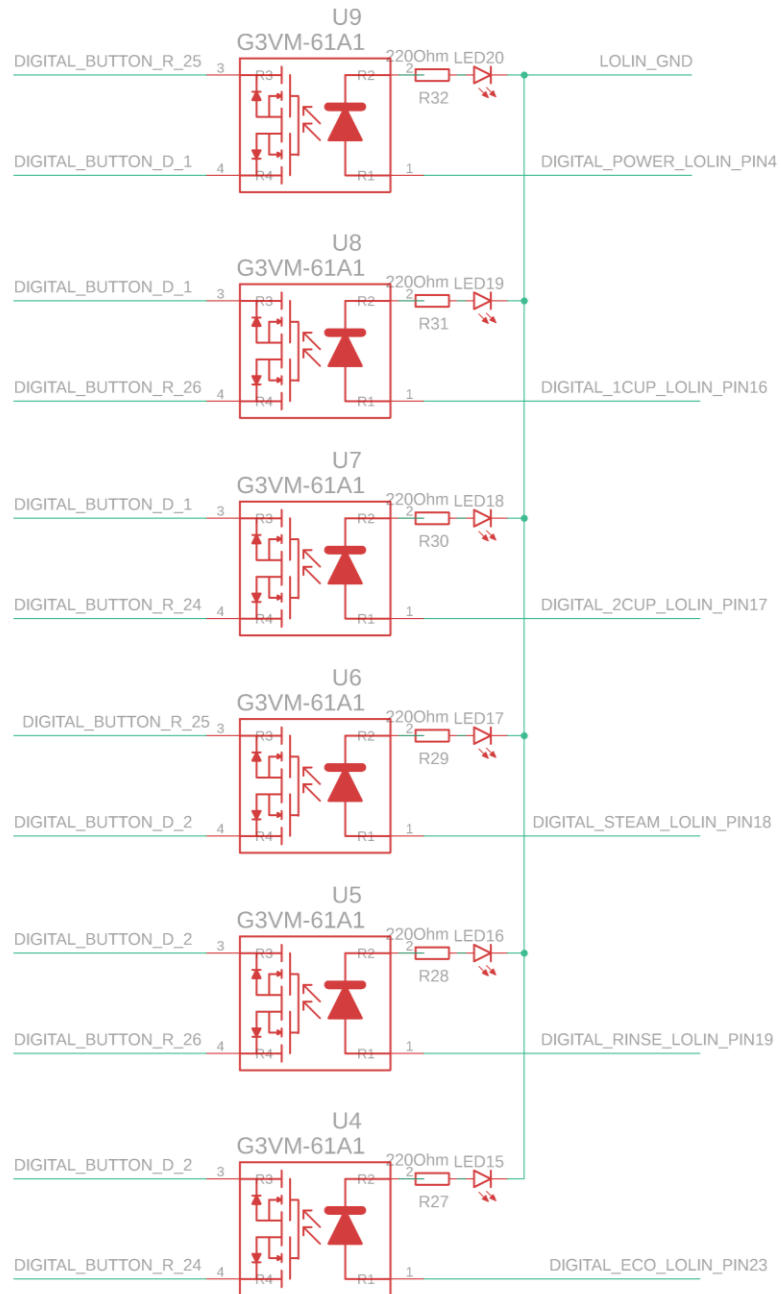
Physical Buttons
Optocoupler connection



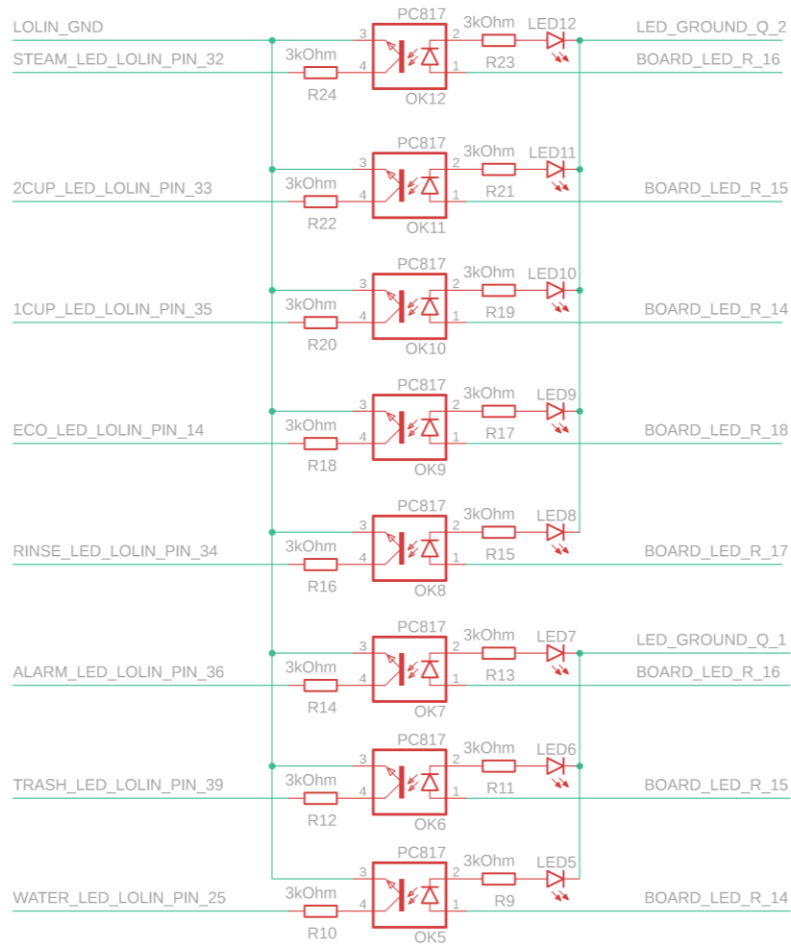
Steuerplatine DeLonghi ESAM Kaffeevollautomat

COFFEE_GND	GND	GND	B_P	BUTTON_POWER
COFFEE_5V	VCC	VCC	B_1C	BUTTON_1CUP
POTI_WIPER_INTENSITY	POTI_A1	POTI_A1	B_2C	BUTTON_2CUP
INTENSITY_POTI_WIPER_BOARD	POTI_A2	POTI_A2	B_S	BUTTON_STEAM
POTI_WIPER_AMOUNT	POTI_B1	POTI_B1	B_E	BUTTON_ECO
AMOUNT_POTI_WIPER_BOARD	POTI_B2	POTI_B2	B_R	BUTTON_RINSE
DIGITAL_BUTTON_R_24	R_24	R_24	R_14	BOARD_LED_R_14
DIGITAL_BUTTON_R_25	R_25	R_25	R_15	BOARD_LED_R_15
DIGITAL_BUTTON_R_26	R_26	R_26	R_16	BOARD_LED_R_16
DIGITAL_BUTTON_D_1	D_1	D_1	R_17	BOARD_LED_R_17
DIGITAL_BUTTON_D_2	D_2	D_2	R_18	BOARD_LED_R_18
LED_GROUND_Q_1	Q_1	Q_1	Q_2	LED_GROUND_Q_2

Digital Buttons
Optocoupler connection



LED Read Board
Optocoupler Connection



5V Powersupply

